BUSINESS REQUIREMENT DOCUMENT (BRD)

Enterprise Aviation Data Platform (EADP)

1. Purpose

This BRD defines the business needs, stakeholder expectations and high-level functional/non-functional requirements for the Enterprise Aviation Data Platform. It is aligned to the aviation capstone problem statement and grounded in the 10 datasets supplied for the project.

2. Business Context

ABC Airways generates data from multiple independent operational and customer-facing systems. The project addresses the need to integrate these heterogeneous datasets, improve trust in the data and provide a common foundation for airline operational and passenger analytics.

3. Business Objectives

ID	Business Objective
BO-01	Centralize aviation data from multiple heterogeneous source systems.
BO-02	Clean, validate and standardize important aviation datasets.
BO-03	Provide a PostgreSQL-based enterprise data platform for curated data.
BO-04	Support flight punctuality and delay analysis.
BO-05	Support aircraft utilization and maintenance planning.
BO-06	Support passenger booking, travel and customer-service analytics.
BO-07	Support airport and baggage operational planning.
BO-08	Enable executive operational and financial reporting.
BO-09	Maintain metadata, audit information and data lineage.
BO-10	Enable collaborative delivery through Git/GitHub.

4. Stakeholder Requirements

Stakeholder	Requirement
Flight Operations	Ability to analyze flight activity, punctuality and delay patterns.
Airport Operations	Ability to use airport master/operational information for planning and analysis.
Customer Service	Ability to analyze customer issues, feedback, support channels and resolution performance.
Maintenance	Ability to use maintenance-task and aircraft-condition information for planning.
Commercial	Ability to analyze booking, ticketing, fare, payment and passenger patterns.
Executive Management	Ability to consume consolidated operational and financial reporting.

5. Functional Requirements

ID	Requirement
FR-01	The platform shall ingest aviation data from the identified source systems.
FR-02	The ingestion solution shall support the source formats required by the assignment, including CSV, Excel, JSON, XML and SQL where applicable.
FR-03	The platform shall load raw source data into PostgreSQL staging tables.
FR-04	The platform shall preserve source identifiers and important source attributes needed for traceability.
FR-05	The platform shall support validation of incoming records and capture ingestion exceptions.
FR-06	The platform shall support standardization of passenger, airport, aircraft and flight-related data.
FR-07	The platform shall support curated datasets suitable for operational and passenger analytics.
FR-08	The platform shall support analysis of flight delays, cancellations and delay reasons.
FR-09	The platform shall support analysis of baggage mishandling and baggage volumes.
FR-10	The platform shall support aircraft maintenance and aircraft sensor information.
FR-11	The platform shall support crew scheduling

	and utilization information.
FR-12	The platform shall support ticketing, booking, payment and cancellation information.
FR-13	The platform shall support CRM/customer-service and loyalty information.
FR-14	The platform shall support airport master and facility information.
FR-15	The platform shall support reporting/analytics through Power BI in the overall solution.

6. Non-Functional Requirements

ID	Requirement
NFR-01	Data quality: ingestion and transformation should include validation and documented quality checks.
NFR-02	Traceability: source records and transformations should be traceable through metadata/lineage mechanisms as the platform evolves.
NFR-03	Maintainability: ETL pipelines should be modular and understandable.
NFR-04	Reliability: ETL processing should provide logging and exception handling.
NFR-05	Scalability: the platform design should accommodate large datasets such as the 70,000-row delay JSON and 80,000-row IoT JSON supplied for this project.
NFR-06	Collaboration: code and documentation should be version-controlled using Git/GitHub.
NFR-07	Consistency: common identifiers, dates, categories and naming conventions should be standardized where appropriate.

7. Data Source Inventory — Project Dataset Basis

File	Mapped Source System	Format	Rows	Structure	Business Use
Flight Operations System.csv	Flight Operations System	CSV	10683	10 columns	Flight operations / route and fare-oriented operational records

Passenger Reservation System.csv	Passenger Reservation System (PRS)	CSV	10683	11 columns	Passenger journey/reservation-oriented records
Airport Operations.csv	Airport Operations Database	CSV	19426	93 columns	Airport master and operational facility information
Baggage Handling System.csv	Baggage Handling System	CSV	101	14 columns	Baggage performance and mishandling analysis
Aircraft Maintenance System.xlsx	Aircraft Maintenance System	Excel	116879	Multiple sheets; Tasks has 18 columns	Aircraft maintenance planning and resource information
Crew_Scheduling_System.xlsx	Crew Scheduling System	Excel	50000	21 columns	Crew scheduling and utilization information
Ticketing_Booking_System.xlsx	Ticketing & Booking System	Excel	60000	25 columns	Ticketing, booking and commercial transaction information
CRM.xlsx	Customer Relationship Management (CRM)	Excel	50000	22 columns	Customer service, loyalty and passenger relationship information
Flight_Delay_Dataset_70000_Rows(1).json	Flight Delay Reports	JSON	70000	22 fields	Flight punctuality and delay analysis
Aircraft_IoT_Sensor_Logs_80000_Rows(1).json	Aircraft IoT Sensor Logs	JSON	80000	20 fields	Aircraft condition monitoring and sensor analytics

8. Dataset-Specific Business Requirements

- Flight Operations: use journey, route, timing, duration, stops and price-related operational records.
- Passenger Reservation System: provide reservation/journey information for passenger travel analysis.
- Airport Operations: provide airport identity, location, facility, service and status information.
- Baggage Handling: provide carrier, passenger, baggage volume and mishandled-baggage measures.
- Aircraft Maintenance: provide maintenance tasks, intervals, execution history and resource-supporting information.
- Crew Scheduling: provide crew assignments, duty/flight/rest hours, licensing and duty status information.
- Ticketing & Booking: provide booking, fare, payment, cancellation, refund, seat and baggage-allowance information.
- CRM: provide customer profile, loyalty, spend, feedback, issue, support and resolution information.
- Flight Delay Reports: provide departure/arrival delay measures and delay reasons, including cancellation/diversion indicators.
- Aircraft IoT: provide aircraft sensor readings, location, fuel, engine, hydraulic, cabin and sensor-status information.

9. Business Rules / Quality Expectations

- Source identifiers should be retained during ingestion whenever present.
- Date and time fields should be converted to consistent database-compatible types.
- Numeric measures should be validated for type and reasonable domain constraints before downstream use.
- Duplicate records should be identified during data-quality activities.
- Missing or invalid values should be documented and handled according to the transformation rule defined for each source.
- Rejected records or ingestion failures should be logged rather than silently discarded.

10. Scope for Sprint 0 and Sprint 1

Sprint 0 establishes the business requirements, architecture, project governance and backlog. Sprint 1 establishes source discovery and ingestion into PostgreSQL staging. Full data profiling, dimensional warehouse construction, lineage implementation and final analytics are later roadmap activities in the supplied assignment and are not required as current execution deliverables.

11. Acceptance Criteria

- Business requirements are traceable to the aviation problem statement.
- All 10 project datasets are included in the source inventory.
- Each dataset has an identified source-system mapping and business purpose.
- Architecture supports the stated technology stack.
- Sprint backlog contains the work needed to achieve Sprint 0 objectives.
- Sprint 1 requirements are clearly reflected in the ingestion-related requirements.

12. Requirement Traceability Summary

Business Objective	Key Requirements	Architecture Component
BO-01 Centralize data	FR-01 to FR-04	Source + Pentaho + PostgreSQL staging
BO-02 Improve data quality	FR-05/FR-06 + NFR-01/NFR-07	Python/Pandas + transformation layer
BO-03 Enterprise storage	FR-03/FR-07	PostgreSQL
BO-04 Flight analytics	FR-08	Curated warehouse + Power BI
BO-05 Maintenance analytics	FR-10	Maintenance + IoT data pipelines
BO-06 Passenger analytics	FR-07/FR-12/FR-13	PRS + Ticketing + CRM + Power BI
BO-09 Traceability	NFR-02	Metadata / audit / lineage
BO-10 Collaboration	NFR-06	Git / GitHub